

Maximization of a Linear Function of Variables Subject to Linear Inequalities

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The general problem indicated in the title is easily transformed, by any one of several methods, to one which maximizes a linear form of non-negative variables subject to a system of linear equalities. For example, consider the linear inequality $ax + by + c < 0$. The linear inequality can be replaced by a linear equality in nonnegative variables by writing, instead $a(x_1 - x_2) + b(y_1 - y_2) + c - z = 0$, where $x_1 \geq 0$, $x_2 \geq 0$, $y_1 \geq 0$, $y_2 \geq 0$, $z \geq 0$. The basic problem throughout this chapter will be consider in the following form:

Problem: Find the vlaues of $\lambda_1, \lambda_2, \dots, \lambda_n$ which maximize the linear form

$$(1) \quad \lambda_1 c_1 + \lambda_2 c_2 + \dots + \lambda_n c_n$$

subject to the conditions that

$$(2) \quad \lambda_j \geq 0 \quad (j = 1, 2, \dots, n)$$

and

$$(3) \quad \begin{aligned} \lambda_1 a_{11} + \lambda_2 a_{12} + \dots + \lambda_n a_{1n} &= b_1, \\ \lambda_1 a_{21} + \lambda_2 a_{22} + \dots + \lambda_n a_{2n} &= b_2, \\ &\vdots \\ \lambda_1 a_{m1} + \lambda_2 a_{m2} + \dots + \lambda_n a_{mn} &= b_m \end{aligned}$$